

THIAGO SAUD

SENIOR AI/MLOPS ENGINEER | SENIOR FRONT-END ENGINEER

Rio de Janeiro, BR | thiago@saudtechnology.com | <https://saudtechnology.com> | [linkedin.com/in/thiagosaud](https://www.linkedin.com/in/thiagosaud)

PROFESSIONAL SUMMARY

Software Engineer with +7 years of experience in mission-critical web platforms and regulated financial ecosystems. Reports to C-Level executives conducting Code Comment Requests (RFCs), Design Architecture Documents (ADRs), Achievement Document (Brag Document) and Roadmaps in front of squads of up to 5 engineers. Specialist in Product Architecture, Front-End, AI/MLOps, Computer Vision, Facial Recognition/Detection, Augmented Reality (AR) and Virtual (VR). In continuous specialization in AI/MLOps engineering (Deep Learning, Machine Learning, RAG, GenAI, Autonomous Agents, LangChain/LangGraph and MLflow). It designed +10 systems in production, including at BTG Pactual Bank, which manages more than R\$ 1.6T+ in assets and serves 4.8M+ customers, contributing with part of the revenue growth of 37% and a ROAE of 28.1% (3Q25); and at XP Investments with R\$ 1.2T+ in assets and 4.5M+ of customers (4Q22). Reduced delivery time by about 65% with AI-assisted workflows and shared libraries and raised code quality and security by approximately ~80%—90% using LLMs/VLMs, MCP, Cursor/Copilot, Prompt/Context/Harness Engineering and Guardrails. Led the implementation of Corporate Design System, Engineering Governance and Software Architecture, increasing consistency and delivery speed.

WORK EXPERIENCES

PRINCIPAL FRONT-END ENGINEER | BTG Pactual Bank S.A.

FEB/2023 — JUN/2026

Digital Equities · São Paulo, BR (Remote) · Direct report to the Managing Partner and the Executive Director · Mentoring of 2 to 5 engineers

Served as a Specialist in one of the largest banks in Latin America, in the Digital Equities vertical, with direct reporting to the Managing Partner and the Executive Director. He led engineering governance, including the preparation of RFCs, ADRs and Achievement Document. Mentored up to 5 engineers, promoting professional development and technical excellence; Conducted the adoption of Artificial Intelligence (AI) Engineering, using LLMs/VLMS, with tools such as MCP, Cursor/Copilot, Prompt/Context/Harness Engineering, Skills and Calling Tools, generating a significant reduction of ~65% in delivery time and gain of ~80% in code quality. ; Contributed with part of the service of 4.8M+ active customers and management of R\$1.6T+ in AUM/Custody, sustaining high-frequency and low latency operations in B3, which boosted the growth of 37% YoY in revenue, ROAE of 28.1% and the historical record of Cost-to-Income of 34.1% (3Q25).

Strategic Products:

- **Home Broker Web (White-Label):** Led the project, coordination, and execution of the design for a digital trading platform serving over 4.8 million active clients. The platform was optimized for high-frequency, real-time trading on the B3 exchange and developed in compliance with rigorous regulatory standards.
- **COE (Structured Operations Certificate):** Worked on the design and implementation of a platform for complex fixed-income and derivative financial products. The platform utilizes a micro-frontend architecture integrated with advanced authentication and regulatory compliance workflows.
- **Internal Corporate Library:** Responsible for the design and governance of a library of reusable, standardized Angular components for the entire Digital Equities vertical, contributing to optimized operational efficiency and cost reduction for the financial institution.

Architecture, Engineering and Artificial Intelligence:

- Accelerated delivery by ~65% and improved code quality/security by ~80–90% by leading the adoption of GenAI (LLMs, VLMs, MCP, prompt/context engineering, and tool calling) and a corporate design system (ORQUESTRA), while also integrating an internal corporate library into the vertical's development workflow—ensuring compliance with WCAG/A11Y standards and Storybook, and managing private package distribution via Nx/Turborepo and Nexus.
- Designed orchestration layers using Angular Signals/Zoneless, RxJS, SOLID, DDD, Clean Architecture, style guides, and AWS S3/CloudFront, isolating business logic and reducing CPU usage and infrastructure costs in a highly volatile environment.
- Implemented distributed micro-frontends using Webpack Module Federation and established continuous observability (Datadog + Sentry) alongside DevOps and CI/CD guardrails (Azure DevOps, GitHub Actions, SonarQube, Playwright, Karma), enhancing stability and compliance with the OWASP Top 10.

Senior Front-End Engineer | XP Investments S.A.

JUL/2021 — NOV/2022

Global Markets · São Paulo, BR (Remote) · Report to the Engineering Manager and the Director of Operations/CFO/Treasury

Served as a Specialist at Latin America's largest investment brokerage within the Global Markets division, reporting to the Engineering Manager (EM) and the Director of Operations/CFO/Treasury. Developed platforms for the trading desk used by 12,000 investment advisors serving approximately 4.5M+ clients. Collaborated strategically with Sales, Advisor, Design, and Backend teams, contributing to the governance of platforms that manage R\$1.2T+ in total assets and drive R\$19.8B in gross revenue (4Q22).

Strategic Products:

- **Black Sales (Trading Desk):** Designed and implemented the registration and distribution platform for structured products and derivatives, supporting the management of +R\$1.2T in total assets.
- **Black AAI (Advisors):** Designed and implemented the profile analysis, recommendation, and portfolio management platform used by approximately 12,000 advisors to serve around 3.8 million active clients.

Architecture, Engineering and Artificial Intelligence:

- Accelerated delivery by ~33% and improved code quality/security by ~80–90% by leading the adoption of a corporate design system (SOMA), ensuring compliance with WCAG/A11Y standards.
- Architected high-performance layers using React, React Hooks, Redux Saga, React Query, GraphQL/Apollo Client, Docker, SOLID, DDD, Clean Architecture, style guides, and AWS S3/CloudFront, isolating business logic and reducing CPU usage and infrastructure costs in a highly volatile environment.
- Implemented distributed micro-frontends using Webpack Module Federation and established continuous observability (Datadog + Sentry) alongside DevOps, guardrails, and CI/CD (Azure DevOps, SonarQube, Playwright, Jest), enhancing stability and compliance with the OWASP Top 10.

Mid-level Front-end Engineer | Zoox Technology Ltda.

JUL/2020 — NOV/2020

Rio de Janeiro, Brazil (Remote) · Technical Lead and Systems Analyst; direct liaison with senior stakeholders

Served as a Technical Reference and Systems Analyst for a highly complex white-label Property Management System (PMS) platform, maintaining direct communication with senior client stakeholders. Integrated real-time billing with global Online Travel Agencies (Booking.com, Expedia, and Agoda), eliminating data inconsistencies and overbooking. Modeled and implemented the end-to-end facial detection and recognition workflow for user validation.

- Developed an asynchronous check-in and check-out workflow—reducing wait times and eliminating bottlenecks—using Vue.js, Vuetify, Docker, and PostgreSQL.
- Designed the end-to-end workflow for facial detection (bounding boxes) and recognition (face matching with confidence scores) for biometric validation, ensuring strict compliance with the National Guest Registration Form (FNHR) and LGPD, utilizing AWS Rekognition and OWASP Top 10 standards.
- Integrated real-time billing with global OTAs (Booking.com, Expedia, and Agoda), eliminating data inconsistencies and overbooking. Architected the in-room monetization module using centralized state management with Vuex.

Junior Front-End Engineer | TG4 Technology Ltda.

SET/2019 — JUL/2020

Rio de Janeiro, Brazil (On-site) · End-to-End Systems Analyst · Direct report to the Co-Founder and CTIO

Served as an end-to-end Systems Analyst, reporting directly to the Co-Founder and CTIO. Planned and implemented a white-label chatbot platform targeting the country's major telecommunications carriers (TIM, Claro, Oi, Vivo, and Nextel). The product integrated predictive recommendation algorithms combined with Natural Language Understanding (NLU), resulting in a 90% increase in the acceptance of upgrade offers, a significant reduction in Customer Acquisition Cost (CAC) and Cost Per Click (CPC), and the maximization of Customer Lifetime Value (LTV).

- Designed orchestration layers using Angular/Zone.js, NGRX/RXJS, Docker, SOLID, Clean Architecture, style guides, and AWS S3/CloudFront/Buckets, isolating business logic and reducing CPU consumption and infrastructure costs in a highly volatile environment.
- Integrated predictive recommendation and classification models focused on Next Best Action and purchase propensity, powered by consumption, billing, and CRM data.
- Established continuous observability using Datadog and Sentry, along with DevOps and guardrails/CI/CD (Azure DevOps, SonarQube, Playwright, Karma), enhancing stability and compliance with the OWASP Top 10.

Technological Research and Development (R&D) Intern | Neoris Brazil Ltda.

APR/2017 — APR/2019

Rio de Janeiro, BR (Hybrid) · Sole R&D intern · Spatial computing, geospatial BI, and AI prototypes (facial recognition, AR/VR)

Served as the sole intern in the Technology R&D engineering division, reporting directly to three Senior Business Managers, with a focus on spatial computing, geospatial Business Intelligence (BI), and cognitive AI prototypes (facial recognition, augmented/virtual reality). Planned and implemented two strategic products for the client Aliansce and two internal proofs of concept (PoCs).

Produtos Estratégicos:

- **Aliansce Web (Multi-floor Mapping and Metrics Analysis):** A framework-free web platform designed for shopping center business intelligence, cross-referencing store revenue metrics on an interactive map using Vanilla JavaScript, the Google Maps API, the Google Drawing API, and SVG.
- **Aliansce Mobile (AR/VR – Real-time visualization of spatial metrics):** Native Android spatial computing application featuring 3D tracking, optical target recognition for overlaying operational indicators in augmented reality, developed using Java, Android Studio, and the Vuforia SDK, with geolocation capabilities.
- **Neoris Mobile (AR/VR – Real-time visualization of spatial metrics):** Prototype of a native Android application that renders 3D dashboards—powered by real-time business intelligence for executive presentations—using Java, Android Studio, the Vuforia SDK, and geolocation.
- **Neoris Web (Visão de IA – Ingestão de Mídia e Classificação Cognitiva):** Web application prototype for object detection and automation, person tracking, demographic classification, and content moderation, using Computer Vision and Artificial Intelligence with vanilla JavaScript and Microsoft Azure Cognitive Services.

RELEVANT PERSONAL PROJECTS IN AI/MLOPS

BYD Machine Learning Case Study (Deep Learning · Transformer) [CLICK HERE TO ACCESS](#)

This project presents a Transformer-based model using TensorFlow and Keras for Shakespeare-style text generation. The complete pipeline includes data preparation, tokenization, training, checkpointing (TensorBoard), generation with sampling controls (Temperature, Top-k), and a Gradio interface. The project employs dependency injection and strict typing, and achieves 100% test coverage using pytest.

BYD Machine Learning Case Study (Machine Learning · Regression) [CLICK HERE TO ACCESS](#)

This project presents simple, multiple, and logistic linear regression pipelines applied to real technical specifications of BYD vehicles. The scikit-learn and pandas libraries are used for data processing. The project includes three PNG charts displaying the data, predictions, fitted curves, and metrics (R^2 and precision). It employs dependency injection and strict typing, and achieves 100% test coverage using pytest.

Weather Forecasting & Restaurant Agent Use Cases (Agentic AI · Agents Built from Scratch) [CLICK HERE TO ACCESS](#)

This project presents autonomous agents implemented from scratch in pure Python, without relying on heavy frameworks like LangChain or LlamaIndex. The Weather Forecasting Agent utilizes the OpenWeather API and features context memory. The Restaurant Agent employs PDF parsing, TF-IDF, and cosine similarity. The Routing Agent uses intent classification and dynamic dispatch. The agents feature a layered architecture comprising Domain, Data, Memory, Processor, and Orchestrator components. The project utilizes dependency injection and strict typing, and achieves 100% test coverage using pytest.

KEY COMPETENCIES

- **Technical Leadership & Processes:** Mentoring · Code Review · Hiring · Cross-functional Collaboration · Request for Comments (RFC) · Architecture Decision Record (ADR) · Internal Portfolio (Brag Document) · Agile (Scrum · Kanban) · Executive Communication (C-Level & Stakeholders).
- **Business Domain (Fintech):** Financial Services · Treasury · Online Trading Platform · Trading.
- **Programming Languages:** Python · TypeScript · JavaScript (ES6+) · Java · C/C++.
- **Software Architecture:** Domain-Driven Design (DDD) · Clean Architecture · SOLID · System Design · Design Systems · Micro-frontends (Module Federation) · Monorepos · BEM Methodology.
- **Artificial Intelligence and Agents:** GenAI · LLMs & VLMs (Local/Cloud) · Claude · Architectural Patterns (RAG) · MCP (Model Context Protocol) · Engineering (Instruction · Prompt · Context · Harness) · Tool Calling & Skills · Autonomous Agents.
- **Artificial Intelligence Engineering:** Machine Learning (ML) · Deep Learning (DL) · PyTorch · TensorFlow · Scikit-learn · Fine-tuning (LoRA · QLoRA) · Orchestration Frameworks (LangGraph · LangChain) · Hugging Face ecosystem.
- **MLOps and AI Infrastructure:** MLflow · AWS SageMaker · Azure ML · Model CI/CD Pipelines · Optimization & Deployment (ONNX) · Hugging Face Hub.
- **Front-End Engineering:** Angular (Signals · NgRx · RxJS · Zoneless) · React (Hooks · Redux · Redux-Saga · Context API) · Vue.js · Next.js · HTML5 · CSS3 · SASS/LESS.
- **Performance and Observability:** Core Web Vitals (LCP · INP · CLS) · RUM/APM (Datadog · Sentry) · GPU Rendering · Virtual/Infinite Scrolling · Lazy Loading · Pagination.
- **Data and APIs:** SQL · PostgreSQL · REST · GraphQL (Apollo Client) · WebSockets · React Query · SSR / SSG / ISR · Data Visualization.
- **UX/UI and Accessibility:** Figma · Storybook · User-Centered Design · Wireframing & Prototyping · WCAG / A11Y · Responsive Design.
- **DevOps, Security and Testing:** Docker · AWS (S3 · CloudFront) · CI/CD (GitHub Actions · Azure DevOps) · OWASP Top 10 · Authentication & Identity (OAuth 2.0 · OIDC · SSO · JWT · 2FA) · TLS · CSP · XSS/CSRF · SonarQube · Playwright · Jest · GitHub Copilot.

EDUCATION

Bachelor of Science in Computer Science | Universidade Veiga de Almeida (UVA) JAN/2014 — FEB/2019

Rio de Janeiro, BR

- **Capstone Project — "Facial Recognition for Public Security":** Real-time prototype using Python and OpenCV, featuring detection (Viola-Jones) and facial recognition (Eigenfaces/PCA & Fisherfaces/LDA); accuracy of ~80–100% and average identification time of ~0.8 seconds.

Executive Program in Business Administration and Management | Universitat de Barcelona (UB) JAN/2018 — NOV/2019

Spain, Catalonia

- Business cases, decision-making, and executive communication for technical leadership.

CERTIFICATIONS AND COURSES

PUCRS — Professional and Emotional Skills (Leandro Karnal)	Completed in AUG/2020
Google — Professional Specialization in UX & UI Design	Completed in DEZ/2024
University of Michigan — Professional Specialization in Python	Completed in AUG/2026
Anthropic — AI Fluency: Framework & Foundations	Completed in AUG/2026
DeepLearning.AI — Professional Specialization in Deep Learning	Completed in AUG/2026
DeepLearning.AI — Professional Specialization in Deep Learning	Completed in AUG/2026
IBM — Professional Specialization in AI Engineering	In progress: goal OCT/2026
IBM — Professional Specialization in RAG & AI Agents	In progress: goal NOV/2026
Duke University — Professional Specialization in MLOps	In progress: goal DEZ/2026

LANGUAGES

Portuguese: Native | English: Technical | Spanish: Technical